MALACKATHON 2026 · UNIVERSITY OF MÁLAGA · BETSSON CHALLENGE

Intelligent Game Validation
Automated casino game validator

Project report of team EscuadronMCP · *Versión en español: Informe-del-proyecto.pdf*

Repository	github.com/EscuadronMCP/BetssonEscuadronMCP
Market	betsson.es (Spanish, EUR) · Casino → Pragmatic Play
Validated games	Sweet Bonanza 2500 and Big Bass Halloween Splash 1000
Result	**PASS** in both scenarios and both games, locally and on GitHub Actions
Delivered version	code `a7f13b1` (validated for real on 15/10/2026)

> **In one sentence:** the program opens the browser, goes to betsson.es, logs in, finds a Pragmatic Play game, plays one real-money spin and delivers a graphical green/red report for every check, **with no human intervention**, looking at the screen the way a player would.

1. How to use it

`uv sync` → copy `.env.example` to `.env` with the two accounts → `uv run python -m betsson_validator run`. In ≈ 70–80 s it writes the report to `reports/<date>/index.html`. Another game: `--game big-bass-halloween-splash-1000`. Also from GitHub: *Actions* → *Validación de juegos* → *Run workflow* (report as an artifact plus a summary on the run page).

2. Summary

- **What it validates:** the two journeys of the challenge. *Completed spin* (funded account): the game really spins, the bet and the currency are the ones shown, and **balance after = balance before − bet + win**, both in the game and on Betsson. *Insufficient balance* (empty account): the spin is rejected, the message appears in Spanish and no balance changes.
- **How:** every check is cross-examined by up to **five witnesses**: game screen, reel motion, Betsson balance, game server and the game's messages to the website. What the player sees is mandatory for a PASS, and any contradiction is a FAIL.
- **Cost:** zero AI calls per run; learning a new game costs ≈ 0.001–0.008 $.
- **Deliberately failing test:** every report includes a check that is made to fail on purpose, to show what a FAIL looks like (section 6).

3. Adapts to different games (same rules, no code)

Each game has a **profile** (HUD texts and what its buttons look like) that a vision AI learns once and that is validated before use. After that everything runs locally without AI. Real screenshots from the final runs of the two games, with very different interfaces:



Sweet Bonanza 2500 · ready, minimum bet €0.20



Sweet Bonanza · spinning



Sweet Bonanza · insufficient balance



Big Bass Halloween Splash 1000 · ready, €0.10



Big Bass · spinning



Big Bass · insufficient balance (message in Spanish)

- If the provider **moves a button or changes a text**, it is found again automatically; if it changes completely, it is relearned once and noted.
- **Another market** = another configuration file. **Another provider** = its configuration, learning its games and, if it changes the bet differently or has its own traffic, a registered adapter. Play'n GO is configured but not

demonstrated (its bet is changed from a separate panel).

4. Price: it is cheap

Item	Cost (measured on 15/10/2026)
Each run of an already learned game	**Zero AI calls**: OCR and vision run locally
Adding a new game	≈ 0.001–0.008 $ (Sweet Bonanza 0.0080 $, Big Bass 0.0011 $)
Repairing a profile if the provider changes its screen	≈ 0.002 $, only when it happens
Money wagered per run	€0.20 (Sweet Bonanza) or €0.10 (Big Bass) in scenario 1; *expected* loss ≈ €0.01 (RTP ≈ 94.5 %)
Infrastructure	Free: GitHub Actions with a self-hosted runner in Spain
Time	≈ 70–80 s for both scenarios in parallel
Report size	< 1.5 MB per game, with screenshots and charts (no video)

Compared with manually checking a new game (we estimate 15–30 minutes of a person's time), the cost is negligible.

5. How it works

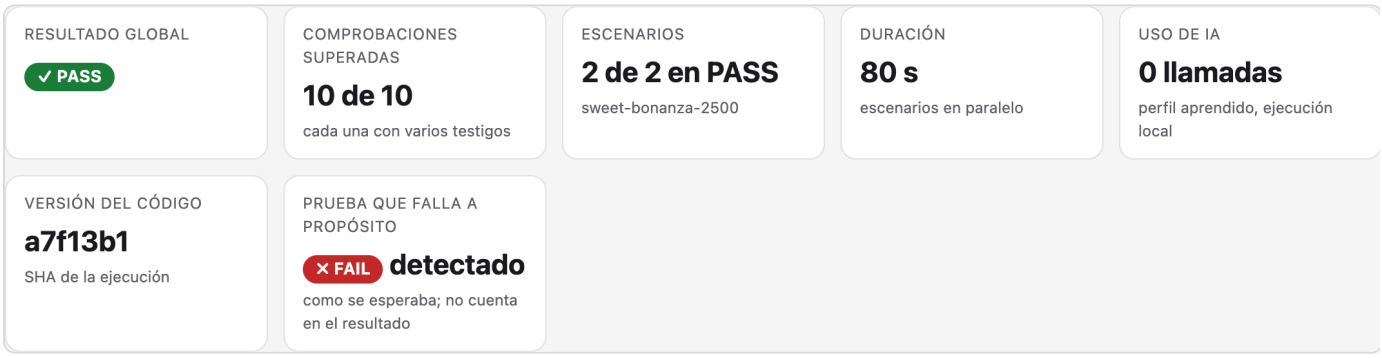
Open Chrome → betsson.es + cookies → Login → Betsson balance → Casino → Pragmatic Play → Game + session → Minimum bet → Preconditions → Observed spin → Round closed → Balances → Report

Witness	What it contributes
Game screen (OCR)	Credit, bet, win, currency and messages the player sees
Reel motion	That the game spun and stopped (clicking the button is not enough)
Betsson header	Website balance before and after
Game server	Accepted bet, declared win and balance, observed without modification and attributed to the game frame
Game messages to the website	`postMessage` events for round, bet, win, balance and language, only from the game frame

Architecture: single-responsibility modules (Betsson website, game, perception, per-provider witnesses in a registry, scenarios, domain, report); dependencies injected behind interfaces; strict typing, linting and **242 tests** in CI. Decisions are documented in `docs/decisiones/` (ADR 001–004).

6. The program's report, in pictures

Screenshots of the report generated by the program in the final Sweet Bonanza 2500 run (the program's interface is in Spanish). The Big Bass one is the same and is in `mentores/`.



Summary panel: result, checks passed, duration, AI usage, code version and the deliberately failing test.

Qué dijo cada testigo

Comprobación	Resultado	Pantalla del juego	Movimiento de los rodillos	Cabecera de Betsson	Servidor del juego	Eve
La tirada empezó y terminó (pulsar el botón no basta).	✓ PASS	✓ confirma jugador	✓ confirma jugador	·	✓ confirma	✓ co
La apuesta mostrada es la apuesta usada en la tirada.	✓ PASS	✓ confirma jugador	·	·	✓ confirma	✓ co
La divisa mostrada coincide con la del mercado.	✓ PASS	✓ confirma jugador	·	✓ confirma	·	✓ co
El saldo del juego es coherente con la tirada completada.	✓ PASS	✓ confirma jugador	·	·	✓ confirma	✓ co
El saldo de Betsson cuadra: después = antes - apuesta + premio.	✓ PASS	✓ confirma	·	✓ confirma jugador	✓ confirma	✓ co

jugador = lo que ve el jugador (pantalla, movimiento o web de Betsson): sin su confirmación no hay PASS. El resto corrobora: si contradice, es FAIL por discrepancia. Pasa el ratón por una celda para ver lo que dijo ese testigo.

What each witness said about each check ("jugador" = what the player sees, mandatory for a PASS).

Valores observados por cada testigo

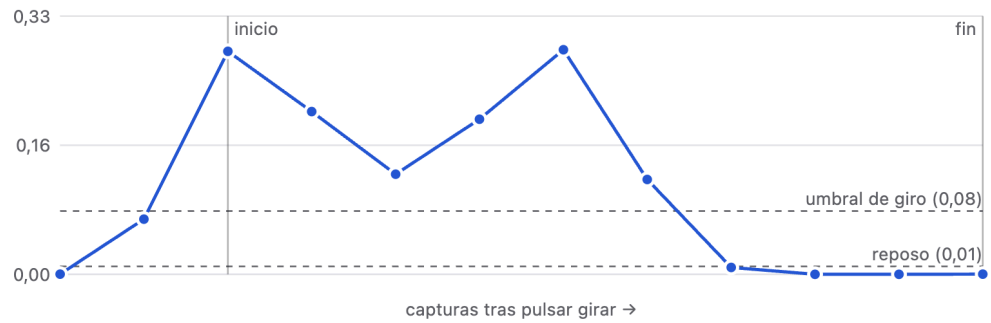
Hecho	Pantalla del juego	Cabecera de Betsson	Servidor	Eventos	
Saldo antes	194,08 €	194,08 €	194,08 €	194,08 €	✓ coinciden
Apuesta	0,20 €	—	0,20 €	0,20 €	✓ coinciden
Premio	0,00 €	—	0,00 €	0,00 €	✓ coinciden
Saldo después	193,88 €	193,88 €	193,88 €	193,88 €	✓ coinciden
Divisa	EUR	EUR	—	EUR	
Idioma	—	—	—	es	

194,08 € − 0,20 € + 0,00 € = 193,88 € — ✓ Betsson muestra 193,88 €

saldo antes − apuesta + premio

Values reported by each witness and the money rule as an equation.

Movimiento de los rodillos



Fracción de la zona de rodillos que cambia entre capturas consecutivas: el juego se movió y volvió al reposo (12 capturas, pico 0.28).

Reel motion chart: the game spun and came back to rest.

Cronología de la ejecución



1. **web abierta** 7.6 s (acumulado 7.6 s)
2. **sesión iniciada** 9.6 s (acumulado 17.2 s)
3. **saldo inicial leído** 0.1 s (acumulado 17.3 s)
4. **juego abierto** 16.3 s (acumulado 33.6 s)
5. **juego listo con apuesta mínima** 15.4 s (acumulado 49.0 s)
6. **giro observado** 7.1 s (acumulado 56.1 s)
7. **ronda cerrada** 0.0 s (acumulado 56.1 s)
8. **marcador estable** 2.0 s (acumulado 58.1 s)
9. **saldo final de Betsson** 4.2 s (acumulado 62.3 s)

Execution timeline: how long each stage took.

🧪 **Deliberately failing test. Every run re-evaluates scenario 1 with what was really observed, but adding €0.01 to the final Betsson balance, as if Betsson credited the spin incorrectly. It does not bet again. The report shows it separately with its FAIL, the reason and what each witness said. It does not count towards the validation result.**

🧪 **Prueba que falla a propósito (demostración; no cuenta en el resultado)**

Prueba que falla a propósito: se vuelve a evaluar este escenario con lo observado de verdad, pero sumando 0.01 € al saldo final de Betsson, como si Betsson abonara mal la tirada. El validador debe detectarlo y marcar FAIL. No apuesta más ni cuenta para el resultado.

✓ El validador detectó el fallo: 1 comprobación(es) en FAIL, como se esperaba.

Comprobación	Resultado	Pantalla del juego	Cabecera de Betsson	Servidor del juego	Eventos del juego
El saldo de Betsson cuadra: después = antes - apuesta + premio.	✕ FAIL	✕ contradice	✕ contradice jugador	✕ contradice	✕ contradice

jugador = lo que ve el jugador (pantalla, movimiento o web de Betsson): sin su confirmación no hay PASS. El resto corrobora: si contradice, es FAIL por discrepancia. Pasa el ratón por una celda para ver lo que dijo ese testigo.

- **El saldo de Betsson cuadra: después = antes - apuesta + premio.** ✕ FAIL: Lo que ve el jugador lo contradice: cabecera de Betsson: saldo 194.08 EUR - apuesta 0.20 + premio 0 = 193.88; Betsson muestra 193.89 EUR.

The deliberately failing test as it appears in the report: the four witnesses contradict the altered balance.



Big Bass Halloween Splash 1000: the same per-witness values and the same money rule, with another game.

7. Security and real money

- **Credentials:** only in `.env` or GitHub Secrets. Login fields are blurred.
- **Evidence scanner:** checks text (including encoded or escaped forms), traces and screenshots read with OCR. It removes any evidence containing a secret, and the report says so.
- **Before spinning:**
  - the bet must be read on screen, in the currency seen, be the game's known minimum and stay below the cap;
  - the funded account must have funds and the "empty" account must not;
  - the credit must match the Betsson balance read at login;
  - bet buttons are only pressed next to the spin button;
  - a spin is never repeated.
- **Accounts:** a per-account lock prevents two runs at once on the same computer. If another session moves money before the spin, it does not spin.

8. Results and verification (code a7f13b1)

Verification	Result
Sweet Bonanza 2500, both scenarios (local)	**PASS** · 70 s · every check confirmed by the available witnesses · deliberate failure detected
Big Bass Halloween Splash 1000, both scenarios (local)	**PASS** · 80 s · all five witnesses confirm every check · deliberate failure detected
GitHub Actions (self-hosted runner in Spain)	**PASS** in both scenarios in 75 s with the final code (run 38775545906), report published as an artifact
Quality	Formatting, linting, typing and 242 tests green in CI
Game accessibility (`--ax-probe`)	Empty tree: the game exposes no semantics, so the screen has to be read

Real failures found during development, and what changed

- **Another session using the same account during a spin.** FAIL by discrepancy. Now there is a per-account lock and, if another session moved money before the spin, it does not spin.
- **Cascade read before the win was collected.** FAIL. Now it waits for the round to close, as signalled by the server and the events.
- **A fake "–" in Big Bass opened the autoplay panel.** The safeguard stopped without spinning. Now the bet is only adjusted next to the spin button and nothing is pressed if it is already the minimum.
- **"Sesión expirada" and "¡Acceda a su cuenta!"** caused by shared accounts. The report quotes the game's notice and saves a screenshot of the failure.

9. Known limitations

- betsson.es only responds from Spain, hence the self-hosted runner. In production it would be an x86 machine in Oracle Cloud Madrid.
- End-to-end compatibility demonstrated: Pragmatic Play (two games). Play'n GO is configured, but its bet adapter is missing.
- The lock does not coordinate different computers. If another session moves money before the spin, it does not spin; if it does so during the spin, the validator flags a discrepancy.
- The optional video is not scanned with OCR (login is blurred at source), so it is not recorded by default.